

Applied Human Computer Interaction (AHCI)

Asif Fazwani

Chapter 5 – Interaction Design Basics

What is Design?

Design = Achieving goals within constraints

- Focuses on purpose and limitations
- Guides decision-making in creating products, systems, or interfaces

Understanding Goals

Key Questions:

What is the purpose of this design? Who is it for? Why do they want it?

Example: Personal Movie Player

- Target users: Young, affluent, mobile users
- Goal: Watch latest movies on the move
- Extra goal: Share experience with friends
- Design starts with understanding **what you are trying to achieve**

What is Design?

Understanding Constraints

Key Questions:

- What materials or standards must we use? Cost limitations?
- Time available? Health and safety considerations?

Example: Personal Movie Player

- Support existing video standards
- Include copyright protection
- Constraints define the **boundaries** within which design happens.

Trade-Offs in Design

Smartphone Design Trade-Off:

Goal:

Slim phone, Long battery life, High performance, Affordable price

Constraint:

Physical battery size, Heat management, Cost of premium components

Trade-Off:

- Thinner phone → Smaller battery
- Bigger battery → Heavier phone

Example: Ultra-slim phones sacrifice battery life.
Gaming phones sacrifice slimness for cooling systems.

The Golden Rule of Design

Understand your materials Before designing anything, you must understand:

- What you are building with
- The strengths and limitations of those materials

Example: AI Model on Cloud vs AI on Device

Same Goal: Face recognition system.

Material 1: Cloud Server

- Massive computing power, Large datasets, High latency possible

Material 2: On-Device (Phone)

- Limited memory, Limited processing, Needs real-time response

The Golden Rule of Design

In HCI, What Are the Materials?

The Computer: Processing power, Memory, Screen size, Sensors, Network speed, Platform constraints

The Human: Attention span, Memory limits, Emotions, Social behavior, Error patterns, Cognitive load

Good design happens when we understand Both.

The Golden Rule of Design

Understanding Humans

Humans: Get distracted easily, Prefer visual content, Have limited working memory, Make mistakes under stress

Example: Why does Instagram widely use:

- Infinite scrolling?
- Visual-first layout?
- Quick reactions (  )?

Because designers understand:

- Attention behavior
- Dopamine loops
- Social validation psychology

The Golden Rule of Design

Understanding Computers

Computers:

- Have processing limits
- Have latency issues
- Can crash
- Can misinterpret inputs
- Need structured instructions

Example: Why do apps show loading spinners? Because:

- Network latency exists
- Processing takes time
- Design must account for technical limits.

The Golden Rule of Design

Example Game designers must understand:

Human:

- Reaction time
- Skill progression
- Frustration tolerance

Computer:

- Frame rate limits
- GPU capacity
- Input delay

Too difficult → User quits

Too easy → User bored

Balance depends on understanding materials.

To Err is Human

People naturally make mistakes. Good design assumes this.

- Humans are not machines
- We forget
- We get distracted
- We act under stress
- We misread things
- Mistakes are normal, not exceptions.

To Err is Human

Human Error – The Misleading Phrase In accidents, reports often say: Human error caused the failure, but often:

- The interface was confusing
- Controls were poorly designed
- Feedback was unclear
- System was inconsistent
- Many “human errors” are actually **design failures**.

Everyday Digital Examples

Example: Messaging Apps

Accidentally sending a message to wrong person.

Solution: Delete for everyone, Message undo option.

Example: Online Payments

Confirmation screen before payment

OTP verification

“Are you sure?” dialog

Because: Users may click accidentally.

High-Stakes Example

In aviation:

- Controls are color-coded
- Critical buttons are separated
- Warnings are redundant (visual + audio)

Why? Because under stress:

- Humans panic
- Reaction time reduces
- Memory weakens
- Good design = error-resistant design.

The Process of Design

Why Process Matters in HCI

Common Industry Problem: Usability treated as  Testing at the end 
Instead of designing from the start

Good Projects/Products do this:

Design usability **from day one**, Users involved early, Iterative improvement

Example: Imagine launching a new campus app

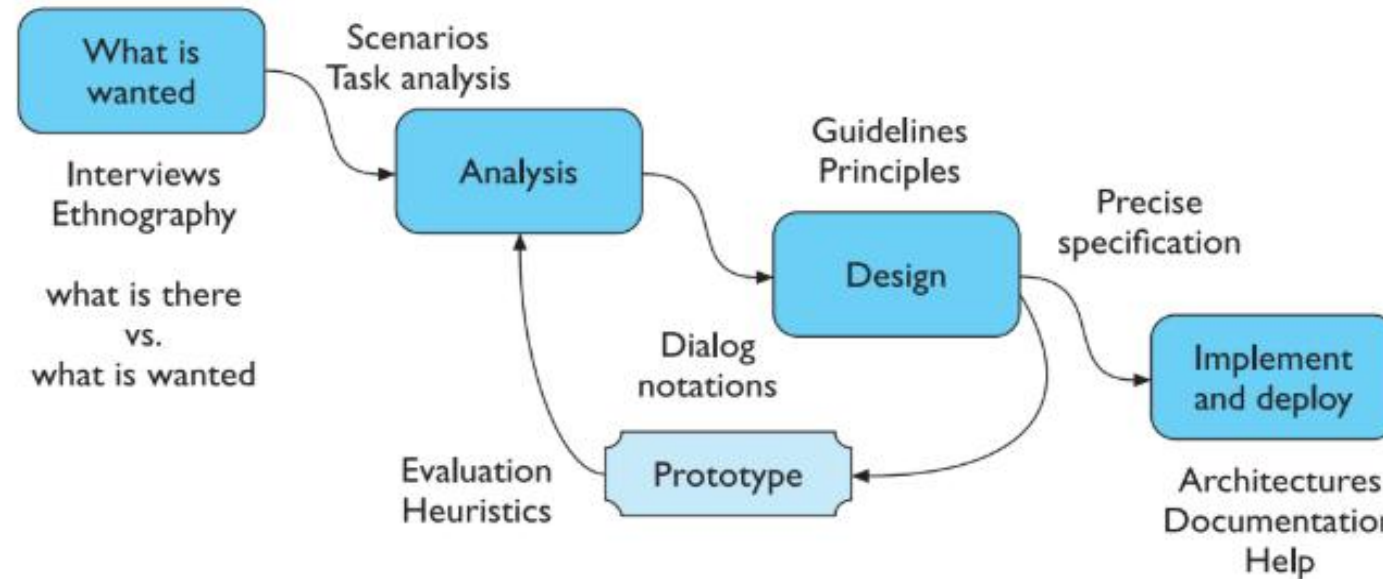
- Students can't find timetable
 - Login keeps failing
 - Buttons are confusing
 - Fixing later = expensive + reputation damage
- Designing right from start = smarter.

Overview of Interaction Design Process

4 Main Phases + Iteration Loop

- Requirements
- Analysis
- Design
- Implementation & Deployment
- Iteration & Prototyping (happens throughout)

It's NOT linear, it's a loop.



Requirements – What is Wanted?

Question: What problem are we solving?

Techniques: Interviews, Observation, Studying existing systems

Example: Designing a New Study Planner App

- How do students currently track assignments?
 - WhatsApp reminders?
 - Notes app?
 - Google Calendar?
- What frustrates them?
 - Forget deadlines
 - Too many notifications
 - Hard to prioritize

You must understand current behavior before building new solution.

Analysis – Making Sense of Data

Now we organize what we learned.

Tools: Task Analysis, Scenarios (interaction stories), User journey mapping

Example: Food Delivery App

Scenario: Saad Jamil is hungry at 1AM. He opens app, searches burger, filters by price, orders in 2 minutes.”

From this:

- Must support fast search
- Simple filters
- Easy payment

Analysis converts messy data into design direction.

Design – From “What” to “How”

Now we decide: Layout, Navigation, Interaction style, Buttons, menus, gestures

Use: Design principles, Guidelines, Heuristics

Example: Designing “Undo Send” in a Messaging App (like WhatsApp)

Design Decisions:

- How long should undo be available? (5 sec? 30 sec?)
- Should it be visible countdown?
- Should receiver see “Message Deleted”?
- Should undo work after message is read?

Trade-off: More privacy for sender **vs** Transparency for receiver

Thinking Time??

If you were redesigning YouTube Shorts to reduce addiction but keep engagement high, what would you change?

Implementation & Deployment

- Write code
- Build hardware
- Prepare documentation
- Launch system

But time and money are limited.

AI Tools in Implementation & Deployment (Speeds Up Development)

Traditional: Developers write every line manually. Slow, repetitive, error-prone

With AI Tools: **ChatGPT** suggests code, **GitHub Copilot** auto-completes blocks, **Cursor** edits inside your IDE

HCI benefit: More resources for usability & interaction focus

Less time buried in boilerplate code → more time designing better interfaces

Iteration & Prototyping

Humans are complex. You will not get it right first time.

So: Build prototype, Test with users, Improve, Repeat

Example: New E-commerce Checkout

Version 1: 5-step checkout → users drop off

Version 2: 1-page checkout → better

Version 3: Auto-fill address → even better

Small improvements → huge impact

Trade Off

- limited time $\Rightarrow$ design trade-off
- usability?
 - finding problems and fixing them?
 - deciding what to fix?
- a perfect system is badly designed
 - too good $\Rightarrow$ too much effort in design

USER FOCUS

Know Your Users: You cannot design good interaction without understanding the people using it.

Stock Control System: Warehouse manager checks inventory.

But indirectly involved:

- Salesperson
- Client
- Auditors
- Assistant manager
- System affects many people.

It's rarely Just ONE User

Even if one person is clicking the screen, there are **many stakeholders**.

Stakeholders

Stakeholders = Anyone affected by the system

Direct users:

- The person clicking

Indirect users:

- People affected by output
- Decision-makers
- Support staff

Who are the stakeholders in a food delivery app?

Who Are Your Users?

Ask: Age? Tech-savvy or novice? Motivation? Goals? Environment? Accessibility needs?

Danger: Designing for a “generic user” There is no such thing.

Better: Create specific user types.

Biggest design mistake: Designing for yourself. Developers often say: It’s obvious. Obvious to whom?

Example: You design for: 20-year-old urban tech user

But actual users:

- 50-year-old shop owner, Rural user with slow internet, First-time smartphone user

Assumption = design failure

Who Are Your Users?

Talk to Them Ways to understand users:

Interviews, Open discussions, Surveys, Participatory design

Participatory Design Users involved in: Idea stage, Prototype testing, Feedback, Decision-making

University Campus Example Designing library booking app.

Instead of guessing: You sit with students, ask how they book now, ask what frustrates them

Result: Better design + students feel ownership

Not only usable, but used.

Who Are Your Users?

Watch Them: What people SAY ≠ What they DO

Example: Student says: I check notifications only 10 times a day.
Reality: 40+ times.

Example (Walking)

- People think they know how they walk.
- But early photography showed: They were wrong.

When black-belt judo players are asked how they throw an opponent, their explanations do not match what they actually do.

Humans are bad at self-reporting.

Who Are Your Users?

Use Your Imagination: Sometimes you cannot access real users.

Then: Use **personas**.

But, not: If I was the user...**INSTEAD:** What would **THEY** do?”

Persona: Mahnoor – 21 year old, Third-Year Student

- Uses phone 6+ hours/day
- Short attention span
- Gets anxious about deadlines
- Prefers visual cues over text
- Studies at night
- Shares notes in WhatsApp groups

Now ask: How would Mahnoor react to this design?

Much clearer than: How would a generic user react?

Scenarios

Scenarios are stories describing how users interact with a system. They represent realistic situations of system use.

- One of the simplest yet most powerful design tools
- Help designers understand user behavior, goals, and environment

Example Imagine a scenario for Spotify: A student jogging wants to quickly skip songs using voice commands without touching the phone.

Question: When you use an app daily, can you describe your interaction as a story?

Example Scenario (Movie Player Case)

Scenario Summary

- A user downloads a movie and transfers it to a **portable movie player**
- Navigates menu options using **arrow keys**
- Connects via **Bluetooth**
- Watches the movie during lunch and shares it with a friend

HCI Design Insight

- Shows **realistic user actions**
- Reveals **interface interactions (menus, icons, controls)**
- Highlights **context of use** (public space, sharing media)

How Scenarios Help Designers

Benefits

- **Communication:** Easy for designers, developers, and clients to understand
- **Validation:** Can test designs against task models and navigation flows
- **Understanding System Behavior:** Shows how the system responds to user actions

Example Designing the "**Snap Streak**" feature in Snapchat required scenarios like:

- Two friends sending snaps daily to maintain streaks.

Linear Nature of Scenarios

Characteristic: Scenarios are **linear stories**.

Advantages: Easy to understand, Humans naturally think in stories

Limitations: Do not show **alternative paths**, May ignore **user mistakes or unexpected actions**

Example Scenario:

- User taps “**Upload Photo**”
- But reality could be:
- Upload fails due to **poor internet**
- User accidentally taps **Delete**

Example Posting on Instagram:

- User edits photo
- Network disconnects
- Post fails

Scenarios Across the Design Process

Scenarios Can Be Used For

- Understanding **user needs**
- Exploring **design ideas**
- Checking **system feasibility**
- Creating **test cases for usability evaluation**

Example

- Before launching a new feature in **Yango**:
Scenario testing might include:
 - User booking ride
 - Driver canceling
 - User rebooking

Navigation Design

- Design is not only about the computer system, but the entire socio-technical context
- Includes users, environment, tasks, and technology
- Eventually designers must focus on the interface itself

Example When using Microsoft Word, the experience depends on:

- The interface design
- The user's task (writing an assignment)
- The environment (classroom, home, office)

Levels of Interaction

Users interact with systems at **multiple levels**.

1. Widgets: Buttons, menus, icons, Help users perform specific actions

Example: The “**Like**” button in **Instagram** is a widget.

2. Screens or Windows: Organization of elements on a screen, Logical grouping of features

Example: The **Home screen in TikTok** groups videos, comments, and share buttons.

3. Navigation Within the Application: Users move between different sections, Interface should clearly show where the user is

Example: Moving between **Home → Explore → Reels → Profile** in **Instagram**.

4. Environment: Interaction with other systems, Switching between applications, Accessing files or networks

Example: Copying text from **Google Docs** and pasting it into **WhatsApp**.

Why Navigation Design Matters?

Good navigation helps users:

- Know **where they are**
- Understand **what will happen next**
- Move easily between features

Example: In Spotify:

- Bottom navigation bar shows:
 - Home
 - Search
 - Library
- This makes navigation **simple and predictable.**

Two Types of Navigation Structure

1. Local Structure: Navigation within a **single screen or page**; How elements are arranged

Example: The **video page layout in YouTube**

Video, Like button, Comments, Share button

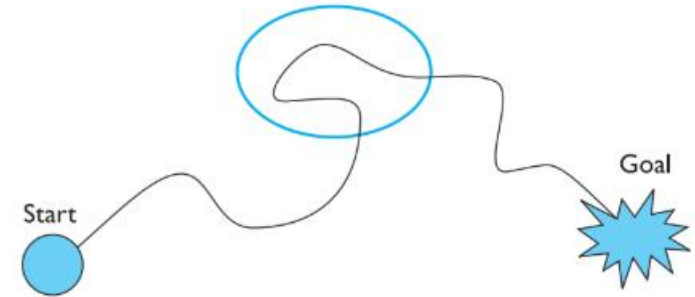
2. Global Structure Overall navigation across the system;
Movement between screens

Example: Global navigation in **Netflix**

Home, TV Shows, Movies, My List

Local Structure in Interaction Design

- Most interaction involves **goal-seeking behavior**
- Users usually have:
 - A **goal in mind**
 - **Partial knowledge** of the system
- Users rarely take the **perfect or shortest path** to their goal
- The figure shows a **user starting at “Start” and moving through a winding path to reach a “Goal”**
- This represents how users **explore interfaces step by step**



Example: Searching for a specific video on **YouTube**:

Open app, Scroll recommended videos, Open one video, Go back, Search manually

The path is **not always direct**.

Designing for Goal-Seeking Users

Users don't need the **most efficient path**, but they must always feel they are **getting closer to their goal**.

Good interfaces help users **understand their progress**.

Example: In **Google Maps**, the navigation system:

- Shows the current position
- Shows the route ahead
- Shows estimated arrival time
- This helps users feel **in control of their journey**.

Four Key Questions in Local Structure

When designing a screen, ensure users can answer four questions:

1. **Where am I?**
2. **What can I do?**
3. **Where will this action take me?**
4. **Where have I been?**

These principles help reduce **confusion and navigation errors**.

Knowing Where You Are

Interfaces should clearly show **the user's location within the system.**

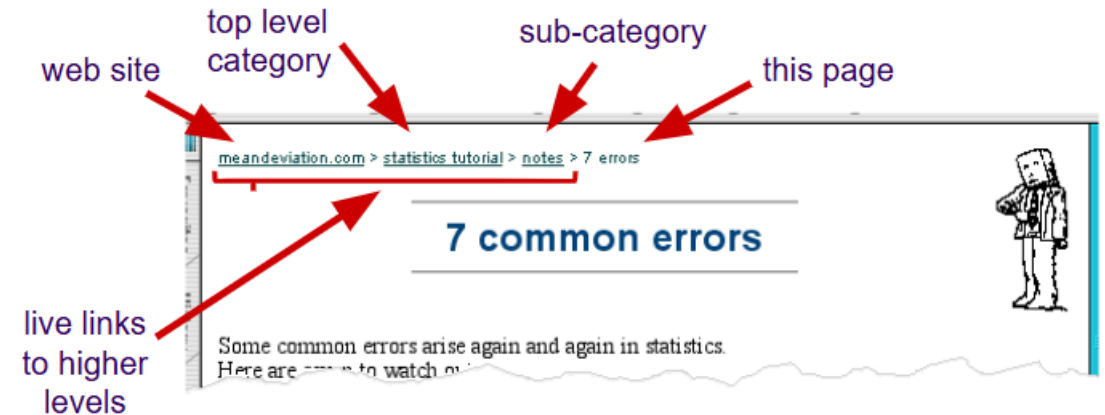
Techniques:

- Page titles
- Breadcrumb navigation
- Highlighted menu items

Example

- In **Amazon Shopping**, breadcrumbs show:
Home → Electronics → Laptops → Gaming Laptops

This tells users **their position in the site hierarchy.**



Knowing What You Can Do

Users must easily identify **interactive elements**.

Good Design

- Buttons look clickable
- Links are clearly marked
- Interactive icons are recognizable

Example The **heart icon in Instagram** clearly indicates: Tap to like a post.

Poor Design Example: Some websites remove **link underlines**, making it unclear what is clickable.

Knowing What Will Happen Next

Users should understand **the result of clicking a button or link.**

Why It Matters: Without this knowledge, users rely on **trial and error**, which causes frustration.

Example

- When booking a ride in **Yango**:
- The “**Confirm Ride**” button clearly indicates what will happen next.

Icons Need Labels

Icons alone can be confusing.

Design Rule

- Icons should be supported by:
- Text labels
- Tooltips
- Explanations

Example: In **Slack**, icons often show tooltips when hovered.

This helps new users **learn the interface quickly**.

Knowing Where You've Been

Users should be able to **track their navigation history**.

Without this, they feel **lost in hyperspace**.

“Lost in Hyperspace”: A term describing **disorientation in complex information systems**.

Example: In Google Chrome:

- Back button
- Browser history
- Recently visited pages

These features help users **recover their navigation path**.

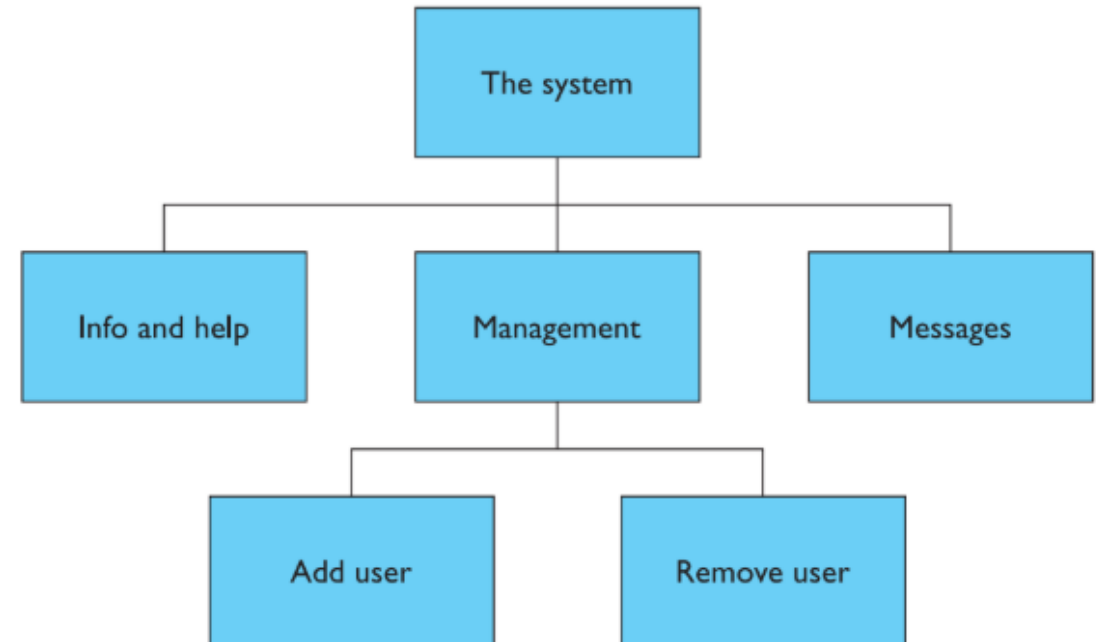
Global Structure in Interaction Design

- **Global structure** refers to the **overall organization of a system**
- Shows how **screens, pages, or device states connect**
- Determines how users **move through the entire application**

Example: In **Instagram**, the main sections include:

- Home
- Explore
- Reels
- Messages
- Profile

This structure helps users **navigate the entire app easily**.



Hierarchical Organization

One common way to organize systems is **hierarchy**.

What is a Hierarchy? A **tree-like structure** where information is grouped into categories and subcategories.

Example:

- Messages
 - ├─ Inbox
 - ├─ Sent
 - ├─ Drafts
 - └─ Spam

Example: Email apps like **Gmail** organize messages using hierarchical folders.

Ways to Organize Hierarchies

Systems can be organized based on:

1. Functional Categories: Group features based on tasks

Example: In **Microsoft Word**: File, Insert, Review, View

2. User Roles: Different users see different structures.

Example: In **Google Classroom**: Teacher view, Student view

3. Modules or Topics: Common in **learning platforms**.

Example: In **Coursera** Course, Module, Lesson, Quiz

Why Hierarchies Works Well

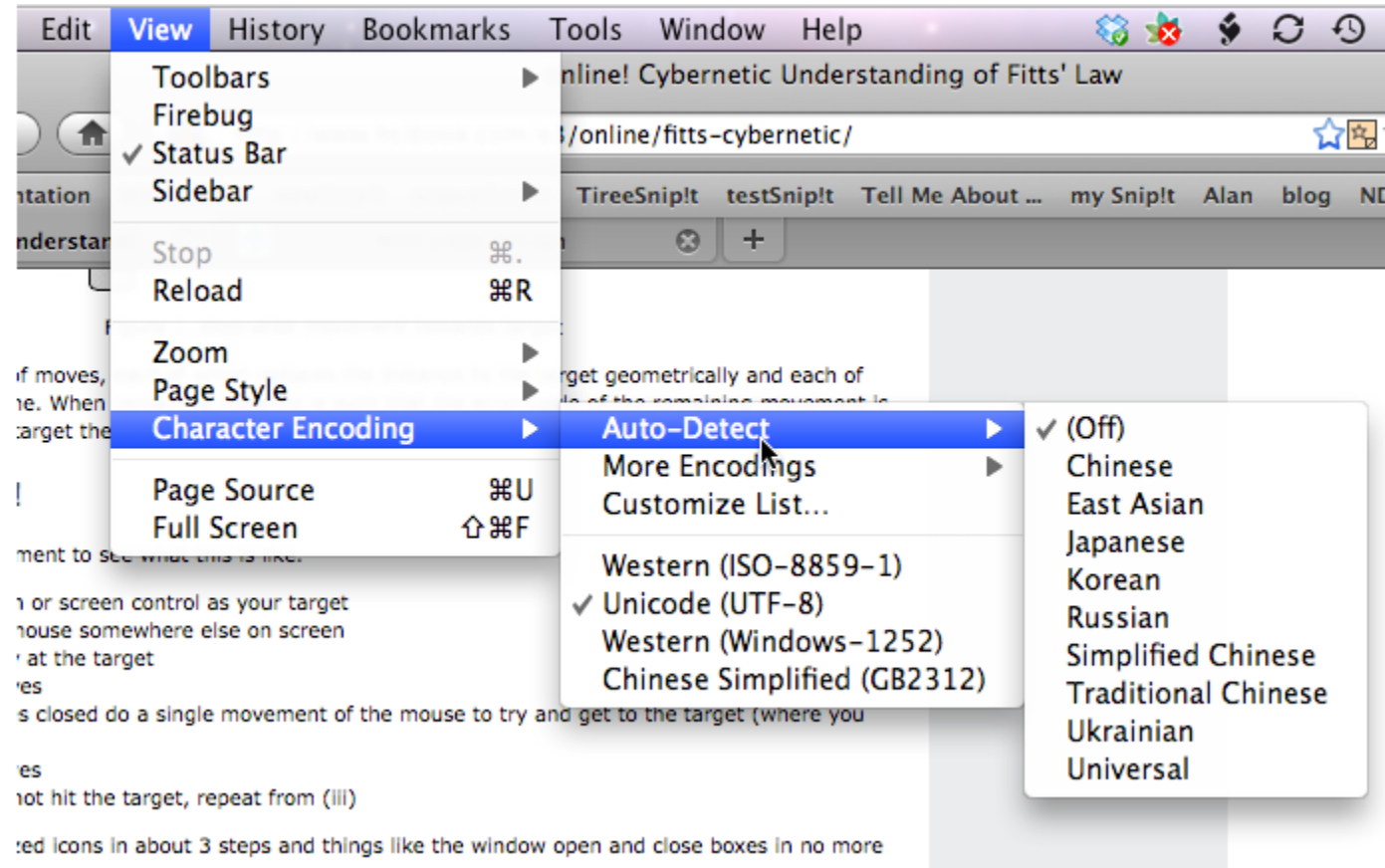
Research shows people often find **hierarchical structures easier to understand**.

Reasons

- Information is **grouped logically**
- Reduces complexity
- Matches how humans organize knowledge

Example

- Shopping websites like **Amazon Shopping** use hierarchical categories:
- Electronics
 - Computers
 - Laptops
 - Gaming Laptops



Deep vs Broad Hierarchies

Deep Hierarchy Many levels of menus.

Example: Menu → Category → Subcategory → Sub-subcategory → Item

Problem: Users may get **lost navigating many levels**.

Broad Hierarchy Fewer levels with **more options per level**.

Example: Category page with many visible options.

Easier for users to **scan quickly**.

Misuse of Miller's Magic Number

Designers often follow **Miller's rule: 7 ± 2 items**.

Reality

- This rule applies to **working memory**, not menu design.
- Menus can contain **many more items** if they are well organized.

Example

- Large directories like **Wikipedia** contain dozens of links but are still usable.

Network Diagrams for Dialog Design

A common way to represent dialog is a **network diagram**.

Network Diagram Shows

- Different **screens or states**
- **Arrows** showing transitions
- **Branches** for choices
- **Loops** for repeated actions

Example:

Main Screen

↙ ↘

Add User Remove User

↓ ↓

Return Confirm Delete

Flow

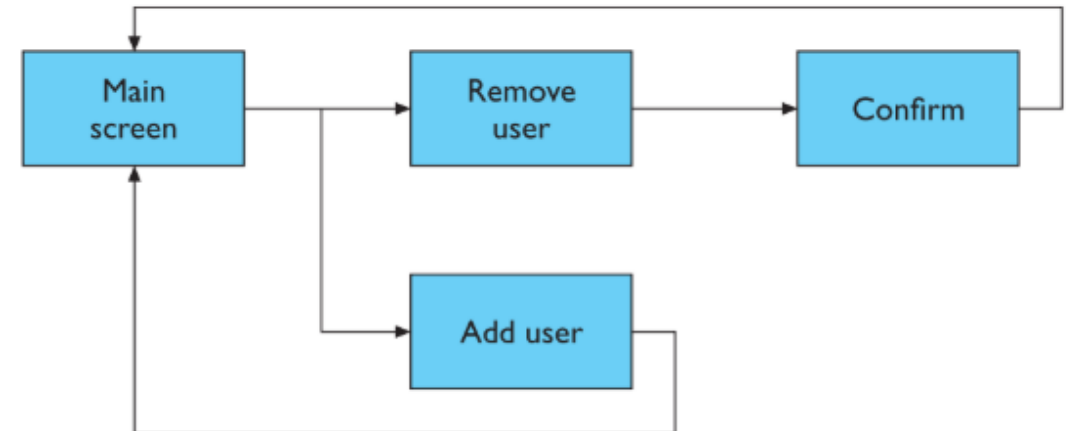
Main Screen → Add User → Return to Main Screen

Main Screen → Remove User → Confirm Removal → Return to Main Screen

Insight

Dialog diagrams show:

- Sequence of actions
- Decision points
- System responses



Benefits of Dialog-Based Design

Dialog structures help designers:

- Understand **task flow**
- Identify **decision points**
- Design **error handling**
 - Improve **user efficiency**

Example

The checkout process in **Foodpanda**:

- Cart → Address → Payment → Order Confirmation
- Each step is part of a **dialog flow**.

Screen Design and Layout

A screen must: **Display information clearly**

Allow users to **interact with the system**

Screen design combines:

- **Psychology (how users perceive information)**
- **Graphic design**
- **Usability principles**

Example: The interface of **Instagram** must

- Show posts clearly
- Allow actions like **like, comment, and share**

Basic Principles of Screen Design

When designing screens, follow three steps:

1. **Ask**

What is the **user trying to do**?

2. **Think**

What information does the user need?

What comparisons might they make?

What order should tasks follow?

3. **Design**

Form follows function

The layout should support the user's task.

Example: In **Yango**, the destination field appears first because the main goal is **booking a ride**.

Tools for Screen Layout

Designers use several **visual tools** to guide how users read and interact with a screen. Main tools include:

- Grouping and structure
- Order of information
- Decoration
- Alignment
- White space

These elements **guide the user's attention**.

Grouping and Structure

Elements that belong together should be **physically grouped together**.

Why? Grouping helps users understand **relationships between information**.

Example: In an online shopping checkout page:

- Order Screen Structure
- Billing Details
- Delivery Details
- Order Items

Billing details: Name Address: ... Credit card no	Delivery details: Name Address: ... Delivery time		
Order details:			
item	quantity	cost/item	cost
size 10 screws (boxes)	7	3.71	25.97
.....	...	...	...

Order of Information

Screens should follow the **natural order of user tasks**.

Example Order

- Enter billing details
- Enter delivery details
- Review items
- Confirm order

Important Rule: The **tab key order** in forms should match the **visual order of fields**.

Example: Forms in **Google Forms** follow logical top-to-bottom input order.

Decoration in Interface Design

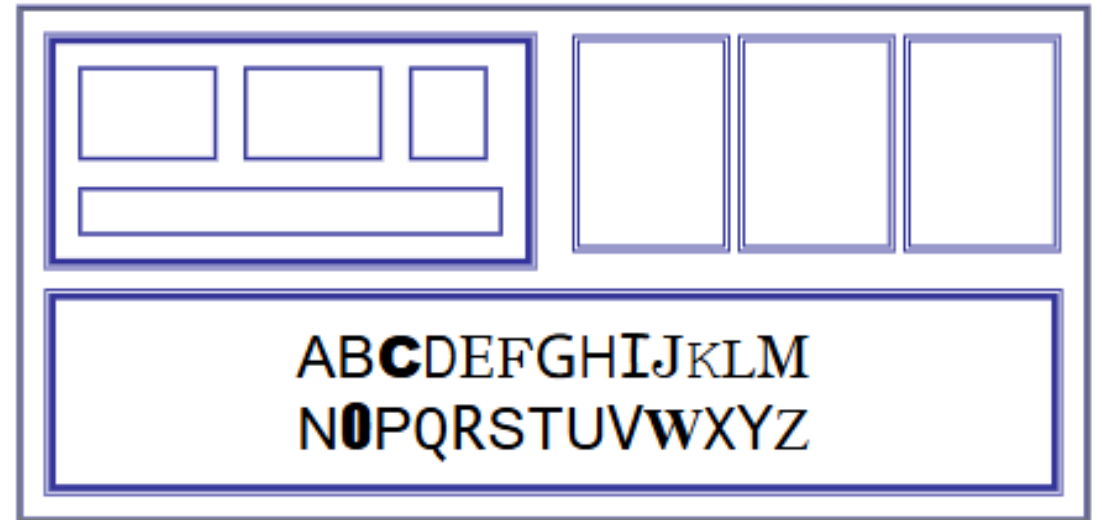
Decorative elements help **visually organize the interface**.

- use boxes to group logical items
- use fonts for emphasis, headings
- but not too many!!

Examples:

- Boxes
- Lines
- Colors
- Fonts

These elements help users identify **different sections**.



Alignment

Proper alignment improves **readability and scanning speed**.

Text Alignment

- Text lists → **Left aligned**
- Numbers → **Right aligned**
- Decimal numbers → **Aligned at decimal point**

Why? Aligned numbers help users quickly compare values.

Example: Financial tables in **Microsoft Excel** align numbers for easy comparison.

Alignment - text

- you read from left to right (English and European)

⇒ align left hand side

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

boring but
readable!

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

fine for special effects
but hard to scan

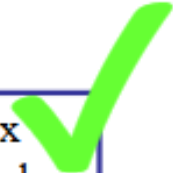
Alignment - names

- Usually scanning for surnames
⇒ make it easy!

Alan Dix
Janet Finlay
Gregory Abowd
Russell Beale



Alan	Dix
Janet	Finlay
Gregory	Abowd
Russell	Beale



Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell



Alignment - numbers

think purpose!

which is biggest?

532.56
179.3
256.317
15
73.948
1035
3.142
497.6256

visually:

long number = big number

align decimal points
or right align integers

627.865
1.005763
382.583
2502.56
432.935
2.0175
652.87
56.34

White Space

White space can create:

- **Separation** Space between sections or paragraphs.
- **Structure** Grouping elements into logical blocks.
- **Highlighting** Drawing attention to key content.

Question Time??

Which app has the easiest sign-up process you've experienced?

User Action and Control in HCI

Interfaces must allow users to **enter information and control the system**

Common interaction areas:

- **Forms**
- **Dialog boxes**
- **Input fields**
- **Selection options**

Poor design makes interfaces **confusing and difficult to use**

If fields are messy or unclear → users abandon registration quickly

Designing Input Fields

- Data entry screens often combine: **Information display and User input fields**
- Key design principles:
 - Maintain **clear alignment**
 - Avoid **jagged layouts**
 - Keep labels **easy to read**

Common Problem: Field labels with different lengths cause **misaligned input boxes**

Design Solutions:

- Use **right-aligned labels**
- Place **labels above input fields**
- Use **smaller fonts for labels**

Example:

- Clean login screen design in Spotify
- Labels placed above fields → easier to scan on mobile devices

Appropriate Appearance in Interface Design

- Interface appearance should match the type of information
- Presentation depends on:
 - Type of data (text, numbers, maps, tables)
 - Technology used (graphics, VR, charts)
 - Purpose of the user task
- **Matching Representation to Purpose:** Different information types need different visual formats.

Better Design Allow sorting by:

Examples:

- **Numbers** → charts, graphs
- **Hierarchies** → tree diagrams
- **Comparisons** → tables
- **Trends** → line graphs

Name

Date

Size

Type

Poor Design Example Alphabetical file list when user wants **recent files**.

Aesthetics vs Utility

A **beautiful interface is not always usable.**

Good design should balance:

- **Visual appeal**
- **Functionality**
- **Clarity**

Benefits of good aesthetics:

- Higher user satisfaction
- Better engagement
- Increased productivity

Background Design and Readability

A common design mistake is **busy backgrounds behind text**.

Problems:

- Low text readability
- Eye strain
- Reduced accessibility

Rule: Text backgrounds should have **low contrast and minimal patterns**.

Problems with Color in Interfaces

Many interfaces misuse color. Common issues:

- Too many colors
- Distracting visuals
- Poor contrast
- Accessibility problems
- **Design Principle** Use color to **support information**, not to **carry all the meaning**.
- **Example:** Like button colors in YouTube are supported by **icons + labels**, not just color.

Problems with 3D Visual Effects

3D visuals can sometimes reduce clarity.

Examples:

- Perspective text is hard to read
- 3D pie charts distort data perception

3D works best for:

- Physical objects
- Spatial data
- Simulations

Localization and Internationalization

Localization: Adapting software for different languages and cultures.

Internationalization: Designing software so it can easily support multiple regions.

Examples of changes:

- Language
- Date formats
- Icons
- Layout direction

Example: Apps like TikTok automatically change:

- Language
- Content
- Cultural references based on region.

Language and Layout Differences

Interface layout depends on **reading direction**.

Examples:

- English → Left to Right
- Arabic → Right to Left

Designers must adjust:

- Menu placement
- Navigation flow
- Layout alignment

Relevant for your region: Apps used in Pakistan often support **English and Urdu interface options**.

Cultural Meaning of Icons

Icons are not always universal.

Example: ✓ Tick vs ✗ Cross

In many **Anglo-American cultures (USA, UK, etc.)**:

- **✓ Tick / Check mark** → means **correct, approved, yes**
- **✗ Cross (X)** → means **wrong, incorrect, no**

Example: Completing a task in Google Docs or confirming an action often uses a **check mark** to show success.

But in some European contexts

- The meanings can be **different or interchangeable**:
- Both ✓ and ✗ may simply be used as **marks** rather than strict “correct vs incorrect”.
- In some grading systems, teachers may mark answers with **X as correct**, which can confuse users from other cultures.